Frozen VLA at Test Time



- Fixed weights (0.4–3.6B params)
- No task-specific fine-tuning
- Variable action quality

Adaptive Conformal Refinement

VLA → K=2 Sample → Adaptive Gate ◇ → CRM → α

615K params | 1.8 MB | ~170ms latency

Selective Improvement

Spatial **+2.1pp** ▲	Goal **+1.3pp** ▲
Object +0.1pp —	Long −0.3pp —

Improves where uncertain, preserves where confident



High uncertainty → CRM active

Gating correctly modulates intervention intensity



Low uncertainty → CRM bypassed

Parameters



615K

parameters (1.8 MB)

Latency



~170ms

all K samples batched

Performance



+2.1pp

Spatial (7.8% trigger)

Validation



2 VLAs

SmolVLA + pi0.5 validated